Report I

Advanced Statistical Mechanics / Statistical Mechanics I

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The final results can be found in (1.7), (2.7), (3.4), (4.9), (5.3), (6.6) and (7.9).

Landau theory

Let us consider the Landau theory:

$$f = t\phi^2 + u\phi^4 - h\phi \qquad (t, h \in \mathbb{R},\ u > 0). \tag{1}$$

(1) In the absence of the magnetic field (i.e., $h = 0$), determine the critical point $t = t_c$.

Solution. The equilibrium value of the order parameter ϕ is determined by the minimum of the free energy, namely:

$$\left.\frac{\partial f}{\partial \phi}\right|_{\phi=\phi_{\mathrm{eq}}} = 0 \tag{1.1}$$

Setting the external field to zero, $h = 0$, we have

$$f(\phi) = t\phi^2 + u\phi^4. \tag{1.2}$$

The extremum condition is

$$\frac{\partial f}{\partial \phi} = 2t\phi + 4u\phi^3 = 2\phi\left(t + 2u\phi^2\right) = 0. \tag{1.3}$$

We then obtain the following solutions

$$\phi = 0, \qquad \phi^2 = -\frac{t}{2u}. \tag{1.4}$$

Using the second derivative

$$\frac{\partial^2 f}{\partial \phi^2} = 2t + 12u\phi^2 \tag{1.5}$$

we discuss the above solutions as follows:

- When $t > 0$, only $\phi = 0$ is a stable minimum.

- When $t < 0$, the following two stable minima appear:

$$\phi = \pm\sqrt{-\frac{t}{2u}} \tag{1.6}$$

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Therefore, the critical point lies at the intersection of the two kinds of behavior,

$$t_c = 0. \tag{1.7}$$

(2) In the absence of the magnetic field, the specific heat c exhibits the following critical phenomenon around the critical point $t = t_c$:

$$c = \begin{cases} a_-|t - t_c|^{-\alpha_-} & (t < t_c), \\ a_+|t - t_c|^{-\alpha_+} & (t > t_c). \end{cases} \tag{2}$$

Calculate the critical exponents $\alpha_\pm$.

Solution. According to the equilibrium order parameter obtained from (1.4), the free energy in equilibrium can be determined as:

- When $t > 0$, the equilibrium point is $\phi = 0$, so

$$f_{\min} = 0. \tag{2.1}$$

- When $t < 0$, the equilibrium point satisfies

$$\phi^2 = -\frac{t}{2u}. \tag{2.2}$$

Substituting back into the free energy,

$$f_{\min} = t\phi^2 + u\phi^4 = t\left(-\frac{t}{2u}\right) + u\left(\frac{t^2}{4u^2}\right) = -\frac{t^2}{4u}. \tag{2.3}$$

Therefore

$$f_{\min}(t) = \begin{cases} 0, & t > 0, \\ -\dfrac{t^2}{4u}, & t < 0. \end{cases} \tag{2.4}$$

According to the thermodynamic relation, the specific heat can be expressed by the second derivative of the free energy with respect to temperature as

$$c = -\frac{\partial^2 f_{\min}}{\partial t^2}. \tag{2.5}$$

Thus

$$c = \begin{cases} 0, & t > 0, \\ \dfrac{1}{2u}, & t < 0. \end{cases} \tag{2.6}$$

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It can be seen that the specific heat does not diverge near the critical point, but only has a finite jump. This is because what is described by (1) is actually a system undergoing a second-order phase transition, and

$$\alpha_+ = \alpha_- = 0. \tag{2.7}$$

(3) The spontaneous magnetization m exhibits the following critical phenomenon around the critical point $t = t_c$:

$$m \propto |t - t_c|^\beta \qquad (t < t_c). \tag{3}$$

Calculate the critical exponent β.

Solution. The spontaneous magnetization m is in fact the value of the order parameter ϕ at equilibrium. According to the solutions of the previous two problems, when $h = 0$ and $t < t_c = 0$,

$$m = \phi_{\text{eq}} = \pm\sqrt{-\frac{t}{2u}}. \tag{3.1}$$

Therefore, its magnitude satisfies

$$|m| \propto |t - t_c|^{1/2}. \tag{3.2}$$

With the definition

$$m \propto |t - t_c|^\beta \qquad (t < t_c) \tag{3.3}$$

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by comparison, we obtain

$$\beta = \frac{1}{2}. \tag{3.4}$$

(4) The zero-field susceptibility χ exhibits the following critical phenomenon around the critical point $t = t_c$:

$$\chi = \begin{cases} c_-|t - t_c|^{-\gamma_-} & (t < t_c), \\ c_+|t - t_c|^{-\gamma_+} & (t > t_c). \end{cases} \tag{4}$$

Calculate the critical exponents $\gamma_\pm$ and the ratio c_+/c_- of the coefficients.

[Note] While the coefficients $c_\pm$ themselves are not universal, their ratio c_+/c_- is universal.

Solution. Now we include the external field and consider the order parameter in equilibrium. Following (1.1), we write down the following equation. In order to keep the notation consistent with the later calculation of the magnetic susceptibility, ϕ_{eq} has already been replaced in advance by m here:

$$h = 2tm + 4um^3. \tag{4.1}$$

According to the definition of the magnetic susceptibility,

$$\chi = \left(\frac{\partial m}{\partial h}\right)_{h\to 0} = \left(\frac{\partial h}{\partial m}\right)^{-1}_{h\to 0} = \left(2t + 12um^2\right)^{-1}\Big|_{h\to 0}. \tag{4.2}$$

Now, using the discussion in Problem 1, we discuss the two cases as follows:

- When $t > 0$, the zero-field equilibrium solution is $m = 0$, so

$$\chi_+ = \frac{1}{2t} = \frac{1}{2}|t - t_c|^{-1}. \tag{4.3}$$

  Therefore

$$\gamma_+ = 1, \qquad c_+ = \frac{1}{2}. \tag{4.4}$$

- When $t < 0$, the zero-field equilibrium solution satisfies

$$m^2 = -\frac{t}{2u}. \tag{4.5}$$

  Substituting this in, we obtain

$$\chi_-^{-1} = 2t + 12u\left(-\frac{t}{2u}\right) = 2t - 6t = -4t = 4|t|. \tag{4.6}$$

  Therefore

$$\chi_- = \frac{1}{4|t|} = \frac{1}{4}|t - t_c|^{-1}. \tag{4.7}$$

  by comparison, we obtain

$$\gamma_- = 1, \qquad c_- = \frac{1}{4}. \tag{4.8}$$

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Thus, the critical exponents $\gamma_\pm$ and the ratio c_+/c_- of the coefficients are:

$$\gamma_\pm = 1, \quad \frac{c_+}{c_-} = \frac{1/2}{1/4} = 2. \tag{4.9}$$

(5) At the critical point $t = t_c$, the magnetization m exhibits the following critical phenomenon:

$$m \propto |h|^{1/\delta}. \tag{5}$$

Calculate the critical exponent δ.

Solution. At the critical point $t = t_c = 0$, (4.1) becomes

$$h = 4um^3. \tag{5.1}$$

Therefore

$$m = \left(\frac{h}{4u}\right)^{1/3} \propto |h|^{1/3}. \tag{5.2}$$

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Comparing with the definition, we obtain

$$\delta = 3. \tag{5.3}$$

Optional assignment

Background Matter

While we have implicitly assumed that ϕ is real, allowing ϕ to take complex values also provides insight into phase transitions and critical phenomena [C. N. Yang and T. D. Lee, Phys. Rev. 87, 404 (1952); T. D. Lee and C. N. Yang, Phys. Rev. 87, 410 (1952)]. Specifically, in contrast with the real-valued regime $\phi \in \mathbb{R}$, the Landau theory f in Eq. (1) generally has three stationary points for the complex-valued regime $\phi \in \mathbb{C}$. For certain parameter choices, two of these stationary points coalesce, signaling a phase transition.

For clarity, let us here assume $t \geq 0$ and $h = i\gamma$ ($\gamma \geq 0$). Then, the stationary solutions for ϕ (i.e., magnetization m) are generally purely imaginary. As described above, two stationary points merge at $\gamma = \gamma_c$, around which unique critical phenomena arise, the Yang-Lee edge singularity [M. E. Fisher, Phys. Rev. Lett. 40, 1610 (1978)].

(6) (optional) Determine γ_c and obtain the following critical exponent Δ:

$$\gamma_c \propto t^{\Delta}. \tag{6}$$

Solution. Setting $\frac{\partial f}{\partial \phi} = 0$, we obtain the following equation:

$$4u\phi^3 + 2t\phi - i\gamma = 0 \tag{6.1}$$

Using the variable substitution $\phi \to i\varphi$, this can be rewritten as the following cubic equation with real coefficients:

$$4u\varphi^3 - 2t\varphi + \gamma = 0 \tag{6.2}$$

Putting it into the standard form:

$$\varphi^3 + p\varphi + q = 0, \qquad p = -\frac{t}{2u},\ q = \frac{\gamma}{4u} \tag{6.3}$$

Using the Cardano discriminant:

$$\Delta_c = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 \tag{6.4}$$

- $\Delta_c > 0$, one real root and two complex conjugate roots
- $\Delta_c = 0$, multiple roots
- $\Delta_c < 0$, all three roots are real

A multiple root is a second-order zero point, so in addition to the first derivative being zero, it also satisfies that the second derivative is zero:

$$\frac{\partial^2 f}{\partial \phi^2} = 2t + 12u\phi^2 = 0 \tag{6.5}$$

This property will be used in the discussion of the next problem.

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Therefore, the condition for having multiple roots is:

$$\gamma_c = \sqrt{\frac{8t^3}{27u}} \sim t^{3/2} \tag{6.6}$$

Therefore:
$$\Delta = \frac{3}{2} \tag{6.7}$$

(7) (optional) Around the critical point $\gamma = \gamma_c$ for given $t > 0$, the (imaginary) magnetization m exhibits the following critical phenomenon:

$$|m - m_c| \propto |\gamma - \gamma_c|^\sigma. \tag{7}$$

Calculate the critical exponent σ.

Solution. Perturb around the equilibrium solution

$$\phi = \phi_c + \delta\phi, \qquad \gamma = \gamma_c + \delta\gamma. \tag{7.1}$$

Substitute this into the equilibrium equation of state (6.1)

$$2t(\phi_c + \delta\phi) + 4u(\phi_c + \delta\phi)^3 - i(\gamma_c + \delta\gamma) = 0. \tag{7.2}$$

Subtract the equation satisfied by the critical point itself, (6.1)

$$2t\phi_c + 4u\phi_c^3 - i\gamma_c = 0, \tag{7.3}$$

we obtain
$$(2t + 12u\phi_c^2)\delta\phi + 12u\phi_c(\delta\phi)^2 + 4u(\delta\phi)^3 - i\delta\gamma = 0. \tag{7.4}$$

The imaginary magnetization we want to discuss corresponds to the double root, and the double root satisfies (6.5)
$$2t + 12u\phi_c^2 = 0, \tag{7.5}$$

so the linear term disappears, leaving only

$$12u\phi_c(\delta\phi)^2 + 4u(\delta\phi)^3 - i\delta\gamma = 0. \tag{7.6}$$

When we are sufficiently close to the critical point, the quadratic term is dominant, therefore

$$\delta\gamma \propto (\delta\phi)^2. \tag{7.7}$$

Thus
$$|m - m_c| = |\phi - \phi_c| \propto |\gamma - \gamma_c|^{1/2}. \tag{7.8}$$

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Therefore
$$\sigma = \frac{1}{2}. \tag{7.9}$$